

Pattern Discovery with Unsupervised Learning

CREDIT CARD CUSTOMER SEGMENTATION
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1. Introduction

The banking industry has experienced a major transformation in recent years due to rapid technological advancements, increasing competition, and changing customer expectations. Modern customers expect banking services that are fast, personalized, secure, and easily accessible through digital platforms. Because of these changes, banks are no longer focused only on providing financial services but also on understanding customer behavior and improving customer experience. In today's highly competitive environment, organizations that effectively analyze customer data are able to make better business decisions, improve customer retention, and gain a stronger competitive advantage in the market. As a result, data analytics and machine learning techniques have become extremely important in the banking sector for identifying customer preferences, understanding spending patterns, and developing personalized services.

Customer segmentation is one of the most widely used analytical techniques in modern banking systems. It refers to the process of dividing customers into different groups based on similarities in their behavior, financial activities, banking habits, and interaction preferences. Through customer segmentation, banks can better understand the diversity that exists within their customer base and design strategies according to the needs of different customer groups. Instead of using a generalized approach for every customer, segmentation helps organizations create personalized marketing campaigns, improve customer satisfaction, enhance service quality, and increase profitability. Effective customer segmentation also supports customer retention by allowing banks to build stronger relationships with customers through targeted communication and customized product offerings.

Traditionally, banks relied mainly on demographic variables such as age, gender, income level, occupation, and geographical location for customer segmentation. However, these traditional methods often fail to capture actual customer behavior and financial interaction patterns. Customers with similar demographic characteristics may still behave very differently in terms of credit card usage, online banking activity, branch visits, and customer service interactions. Therefore, organizations increasingly depend on advanced analytical methods that can uncover hidden patterns within customer data. Machine learning techniques, especially unsupervised learning methods, provide a more efficient and accurate approach for identifying natural groupings among customers based on behavioral similarities rather than predefined assumptions.

Unsupervised learning is a branch of machine learning that analyzes data without requiring labeled outputs or predefined categories. It helps in discovering hidden structures, relationships, and patterns within large datasets. Among various unsupervised learning techniques, clustering algorithms are particularly useful for customer segmentation because they group customers based on similarities in their characteristics and behavior. Clustering techniques enable organizations to identify meaningful customer segments such as high-value customers, digitally active customers, service-dependent customers, and low-engagement customers.

This project focuses on performing customer segmentation for AllLife Bank using unsupervised learning techniques. The project aims to analyze customer behavior and identify meaningful customer groups based on their financial and banking activities. The dataset used in this study includes variables such as average credit limit, total number of credit cards, number of online banking visits, number of physical bank visits, and customer support interactions. By analyzing these features, the project attempts to uncover hidden behavioral patterns that can help the bank improve customer relationship management and business performance.

To achieve the study's objectives, two major clustering techniques, namely K-Means Clustering and Hierarchical Clustering, were applied and compared. K-Means Clustering partitions customers into groups based on similarity, while Hierarchical Clustering helps understand the relationships and hierarchical structure among customer segments. Various evaluation techniques, such as the Elbow Method, Silhouette Score Analysis, and Cophenetic Correlation Coefficient, are used to determine the optimal number of clusters and validate clustering performance. These techniques help ensure that the clusters formed are meaningful, distinct, and useful for business interpretation. Furthermore, customer segmentation can help the bank optimize operational costs by encouraging digitally active customers to use online services while allocating customer support resources more efficiently. Personalized offers and targeted communication strategies can increase customer satisfaction and improve long-term customer loyalty. The insights generated through clustering analysis can therefore support strategic decision-making and contribute to business growth and profitability.

2. Business Problem

2.1 Problem Context

The banking sector is becoming more competitive due to changing customer expectations and increasing digitalization. Customers use banking services in different ways, including online banking, branch visits, and credit card services. Because of this, banks need to better understand customer behavior and preferences. AllLife Bank serves a large number of customers but faces difficulty in identifying meaningful customer groups based on their banking activities. Different customers show different levels of credit card usage, online engagement, and service dependency. AllLife Bank serves a large number of customers but faces difficulty in identifying meaningful customer groups based on their banking activities. Different customers show different levels of credit card usage, online engagement, and service dependency. Currently, the bank uses common business and marketing for all customers, which may not be equally effective for every customer segment. This can affect customer satisfaction, marketing efficiency, and overall business performance. To solve this issue, the bank requires a customer segmentation approach that can group customers based on similar behavior patterns. Therefore, this project uses unsupervised learning techniques to identify different customer segments and generate business insights that can support better decision-making and personalized services.

2.2 Problem Definition

AllLife Bank wants to better understand its customers based on their banking behavior and credit card usage patterns. The bank currently lacks a proper system to identify different customer groups and understand how customers interact with banking services such as online banking, branch visits, and customer support. Different customers show different levels of financial activity and service usage. Some customers actively use multiple credit cards and digital banking services, while others rely more on traditional banking methods. Because of this variation, applying the same marketing and customer engagement strategy to all customers is not effective. The main problem is to identify meaningful customer segments using customer behavioral data. This project aims to apply unsupervised learning techniques, specifically clustering methods, to group customers with similar characteristics and behavior patterns. The identified customer segments can help the bank improve marketing strategies, provide personalized services, increase customer engagement, and support better business decision-making.

2.3 Objectives

The main objective of this project is to analyze customer behavior and identify meaningful customer segments using unsupervised learning techniques.

- To segment customers based on their banking behavior, credit card usage, and interaction patterns.
- To understand the characteristics and preferences of different customer groups for better customer relationship management.
- To help the bank develop targeted marketing strategies and personalized banking services.
- To improve customer engagement, digital banking adoption, and overall business performance through data-driven insights.

3. Data Description

The dataset used in this project contains customer-related information collected from AllLife Bank. The data mainly focuses on customer banking behavior, credit card usage patterns, and interaction activities across different banking channels. The dataset is used to identify meaningful customer segments through clustering techniques. The dataset consists of 660 customer records and includes numerical variables that represent customer financial and banking activities. Each record in the dataset corresponds to an individual customer.

The following variables are included in the dataset:

Variable	Description
Customer Key	Unique Identification number assigned to each customer.
Avg_Credit_Limit	Average credit limit available to the customer.
Total_Credit_Cards	Total number of credit cards owned by the customer.
Total_visits_bank	Total number of visits made to the bank branch.
Total_visits_online	Total number of online banking visits made by the customer.
Total_calls_made	Total number of customer service calls made by the customer.

The variables in the dataset provide insights into customer spending capacity, banking preferences, digital banking usage, and customer interaction behavior. These features are important for identifying similarities and differences among customers and forming meaningful customer segments using unsupervised learning techniques.

4. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the structure of the dataset, identify customer behavior patterns, and examine relationships among different variables. The analysis helped in gaining insights into customer banking activities, credit card usage, and interaction patterns before applying clustering techniques.

4.1 Checking the Structure of the Data

4.1.1 Initial Data Preview

The initial data preview was performed to understand how the dataset is organized and to examine the first few records of the dataset. This step helped in identifying the available variables, understanding the format of the data, and verifying whether the dataset was loaded correctly.

The first five rows are displayed as follows:

SI_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
1	87073	100000	2	1	1	0
2	38414	50000	3	0	10	9
3	17341	50000	7	1	3	4
4	40496	30000	5	1	1	4
5	47437	100000	6	0	12	3

The last five rows are displayed as follows:

SI_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	Total_visits_online	Total_calls_made
656	51108	99000	10	1	10	0
657	60732	84000	10	1	13	2
658	53834	145000	8	1	9	1
659	80655	172000	10	1	15	0
660	80150	167000	9	0	12	2

Observation:

- The dataset contains customer-related numerical data associated with banking behavior and credit card usage.
- Each row represents a unique customer identified using the *Customer Key* variable.
- The dataset includes variables such as average credit limit, total credit cards, bank visits, online visits, and customer support calls.
- Significant variation can be observed in customer banking activities and credit limits.
- Some customers show higher online activity and credit limits, while others show lower banking engagement.
- The variation in customer behavior indicates that the dataset is suitable for customer segmentation using clustering techniques.

4.1.2 Data Shape

(660, 7)

- The dataset contains 660 observations and 7 variables.
- The dataset size is adequate for performing clustering and segmentation analysis.
- Each row represents a unique customer record.
- The dataset includes both identifier and customer behavior-related variables.
- Most of the variables are numerical, which makes the dataset suitable for unsupervised learning techniques.

4.1.3 Viewing Column Names

```
Index(['SI_No', 'Customer Key', 'Avg_Credit_Limit', 'Total_Credit_Cards',  
      'Total_visits_bank', 'Total_visits_online', 'Total_calls_made'],  
      dtype='object')
```

- The dataset contains 7 columns representing customer identification and banking behavior variables.

- Variables such as *Avg_Credit_Limit*, *Total_Credit_Cards*, *Total_visits_bank*, *Total_visits_online*, and *Total_calls_made* are relevant for customer segmentation.
- *Sl_No* and *Customer Key* act as identifier variables and do not contribute directly to clustering.
- The column names are clear, meaningful, and properly formatted for analysis.
- The dataset structure indicates that it is suitable for further preprocessing and exploratory data analysis.

4.1.4 Data Background and Content

#	Column	Non-NULL Count	Dtype
0	Sl_No	660 non-NULL	int64
1	Customer Key	660 non-NULL	int64
2	Avg_Credit_Limit	660 non-NULL	int64
3	Total_Credit_Cards	660 non-NULL	int64
4	Total_visits_bank	660 non-NULL	int64
5	Total_visits_online	660 non-NULL	int64
6	Total_calls_made	660 non-NULL	int64

- The dataset contains 660 entries and 7 columns.
- All columns have **660 non-null values**, indicating that there are no missing values in the dataset.
- All variables have the data type **int64**, which means the dataset consists entirely of numerical variables.
- Since the dataset contains numerical features, it is suitable for clustering and other machine learning techniques.
- Since the dataset contains numerical features, it is suitable for clustering and other machine learning techniques.

4.1.5 Checking NULL Values

```
Sl_No      0
Customer Key  0
Avg_Credit_Limit  0
Total_Credit_Cards  0
Total_visits_bank  0
Total_visits_online  0
Total_calls_made  0
dtype: int64
```

- All variables contain 0 missing values, indicating that the dataset is complete.
- No NULL or missing entries are present in any column of the dataset.
- The absence of missing values reduces the need for additional data cleaning techniques.
- The dataset is clean and suitable for further exploratory data analysis and clustering.

4.2 Checking the Types of Data

```
Sl_No      int64
Customer Key  int64
Avg_Credit_Limit  int64
Total_Credit_Cards  int64
Total_visits_bank  int64
Total_visits_online  int64
Total_calls_made  int64
dtype: object
```

- All variables in the dataset have the data type **int64**, indicating that the dataset contains only numerical values.
- Numerical data types are suitable for applying clustering algorithms such as K-Means and Hierarchical Clustering.
- No categorical variables are present in the dataset.
- The dataset structure is appropriate for statistical analysis and machine learning techniques.
- The *Customer Key* and *Sl_No* columns act as identifier variables, while the remaining columns represent customer behavioral features.

4.3 Checking Statistical Summary

	Sl_No	Customer Key	Avg_Credit_Limit	Total_Credit_Cards	\
count	660.000000	660.000000	660.000000	660.000000	
mean	330.500000	55141.443939	34574.242424	4.706061	
std	190.669872	25627.772200	37625.487804	2.167835	
min	1.000000	11265.000000	3000.000000	1.000000	
25%	165.750000	33825.250000	10000.000000	3.000000	
50%	330.500000	53874.500000	18000.000000	5.000000	
75%	495.250000	77202.500000	48000.000000	6.000000	
max	660.000000	99843.000000	200000.000000	10.000000	

	Total_visits_bank	Total_visits_online	Total_calls_made
count	660.000000	660.000000	660.000000
mean	2.403030	2.606061	3.583333
std	1.631813	2.935724	2.865317
min	0.000000	0.000000	0.000000
25%	1.000000	1.000000	1.000000
50%	2.000000	2.000000	3.000000
75%	4.000000	4.000000	5.000000
max	5.000000	15.000000	10.000000

- The dataset contains 660 records for all variables, indicating complete data availability.
- The average credit limit of customers is approximately 34,574, with values ranging from 3,000 to 200,000, showing significant variation in customer spending capacity.
- Customers own an average of around 4 to 5 credit cards, while the maximum number of credit cards owned is 10.
- The average number of bank visits is around 2, whereas online visits average around 3, indicating moderate digital banking usage among customers.
- Customer support calls range from 0 to 10, with an average of approximately 4 calls.
- Large differences between minimum and maximum values suggest the presence of varying customer behavior patterns within the dataset.
- The wide spread in variables such as credit limit, online visits, and calls made indicates that the dataset is suitable for customer segmentation analysis.

4.4 Checking for Data Irregularities

```
Duplicate values: 0

Negative values in each column:
Sl_No          0
Customer Key   0
Avg_Credit_Limit 0
Total_Credit_Cards 0
Total_visits_bank 0
Total_visits_online 0
Total_calls_made 0
dtype: int64

Customers with Zero Credit Limit: 0
Customers with Zero Credit Cards: 0

Minimum Values:
Sl_No          1
Customer Key   11265
Avg_Credit_Limit 3000
Total_Credit_Cards 1
Total_visits_bank 0
Total_visits_online 0
Total_calls_made 0
dtype: int64
```

- No duplicate values are present in the dataset.
- No negative values are observed in any of the variables.
- There are no customers with zero credit limit or zero credit cards.
- The minimum credit limit observed is 3000, indicating that all customers have active credit limits.
- The minimum number of credit cards is 1, which means every customer owns at least one credit card.
- Some customers have zero bank visits, online visits, or customer service calls, indicating differences in customer interaction behavior.
- The dataset appears clean, valid, and suitable for further analysis and clustering techniques.

4.5 Univariate Analysis

4.5.1 Identifying Numerical & Categorical Variables

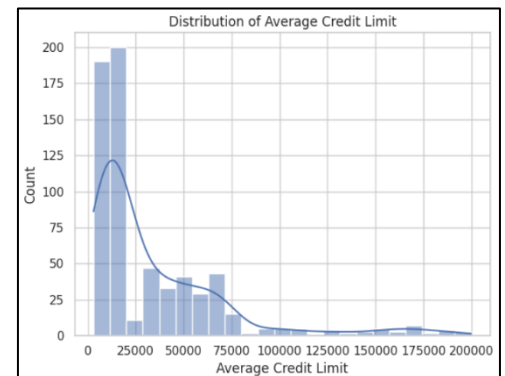
```
Numerical Variables:  
  
Index(['Sl_No', 'Customer Key', 'Avg_Credit_Limit', 'Total_Credit_Cards',  
      'Total_visits_bank', 'Total_visits_online', 'Total_calls_made'],  
      dtype='object')  
  
Categorical Variables:  
  
Index([], dtype='object')
```

- All variables in the dataset are numerical variables.
- No categorical variables are present in the dataset.
- Numerical variables include customer credit limit, number of credit cards, bank visits, online visits, and customer support calls.
- Since the dataset contains only numerical features, it is suitable for clustering and other machine learning techniques.
- Identifier variables such as *Sl_No* and *Customer Key* may not contribute directly to clustering analysis.

4.5.2 Exploring Numerical Variables

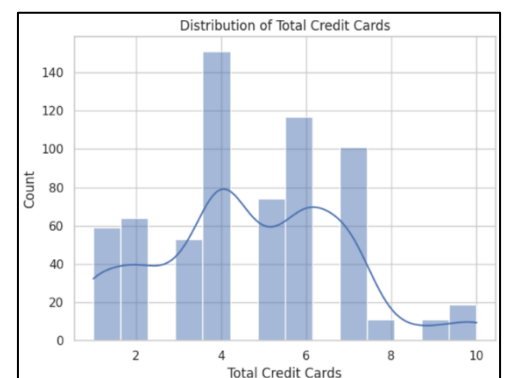
1. Average Credit Limit

- Most customers have average credit limits within the lower to moderate range.
- A smaller number of customers possess very high credit limits compared to the majority of customers.
- The credit limit values show noticeable variation among customers.
- The distribution indicates the presence of customers with different financial capacities and spending behaviors.
- High credit limit customers may represent premium or high-value customer groups.



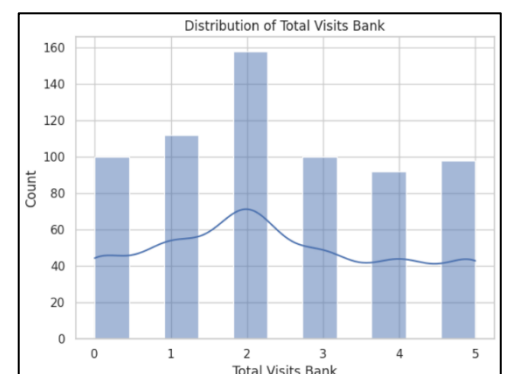
2. Total Credit Cards

- Most customers own between 3 to 7 credit cards.
- A large number of customers possess around 4 to 6 credit cards.
- Very few customers own a high number of credit cards such as 9 or 10.
- The number of credit cards varies across customers, indicating differences in customer financial engagement.
- Customers with higher numbers of credit cards may represent more active or premium banking customers.



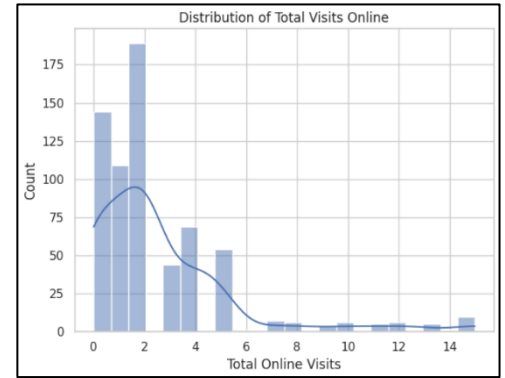
3. Total Bank Visits

- Most customers visit the bank branch between 1 to 3 times.
- The highest number of customers have made around 2 bank visits.
- Some customers do not visit the bank branch at all, indicating reliance on other banking channels.
- A smaller number of customers frequently visit the bank branch up to 5 times.
- The variation in bank visit frequency reflects different customer banking preferences and interaction patterns.



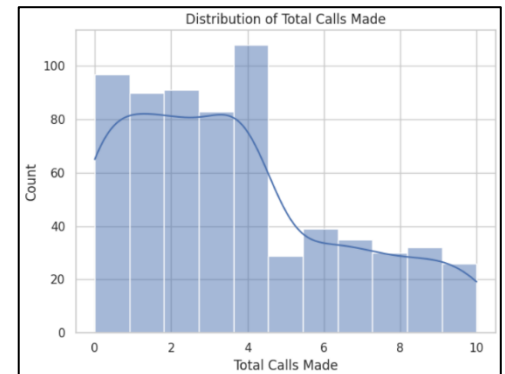
4. Total Online Visits

- Most customers make between 0 to 5 online visits.
- A large number of customers have low to moderate online banking activity.
- Some customers show very high online visit counts, reaching up to 15 visits.
- The presence of customers with high online activity indicates strong digital banking engagement.
- The variation in online visits reflects different levels of customer preference for digital banking services.



5. Total Calls Made

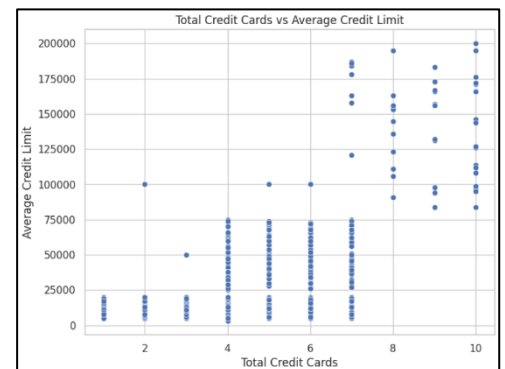
- Most customers make between 0 to 5 customer service calls.
- A large number of customers have moderate interaction with customer support services.
- Some customers make a higher number of calls, reaching up to 10 calls.
- The variation in the number of calls indicates differences in customer service dependency among customers.
- Customers with higher call frequency may require additional support or assistance compared to other customer groups.



4.6 Bivariate Analysis

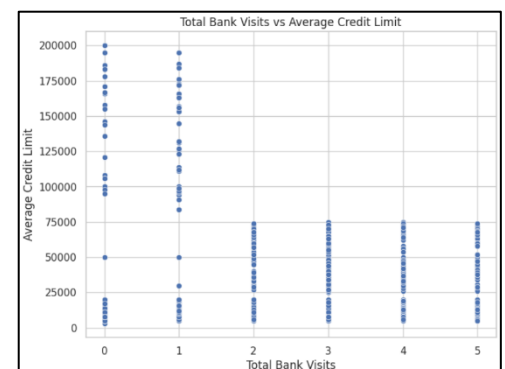
4.6.1 Average Credit Limit vs Total Credit Cards

- Customers with a higher number of credit cards generally tend to have higher average credit limits.
- Customers owning 1 to 3 credit cards mostly have lower credit limits.
- Customers with 7 to 10 credit cards show significantly higher credit limits compared to other customers.
- A noticeable increase in credit limit can be observed as the number of credit cards increases.
- The relationship indicates a positive association between total credit cards and average credit limit.
- High credit card ownership may represent financially active or premium customer groups.



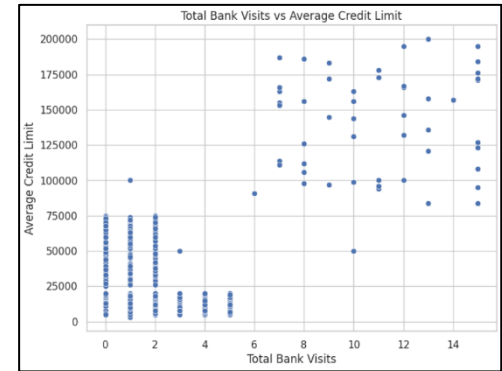
4.6.2 Average Credit Limit vs Total Bank Visits

- Customers with lower bank visits show a wider range of average credit limits.
- Customers making 0 or 1 bank visits include several high credit limit customers.
- Customers with frequent bank visits generally have moderate credit limits.
- High credit limit customers appear to rely less on physical bank visits.
- The relationship suggests that customers with higher financial capacity may prefer alternative banking channels over branch visits.



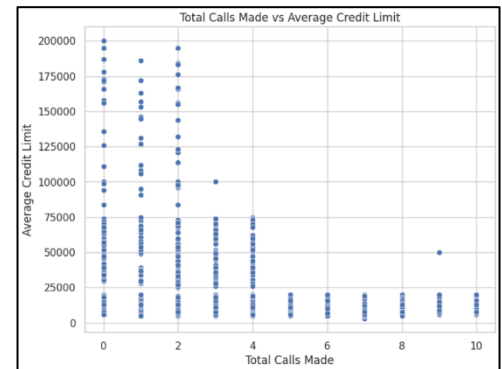
4.6.3 Average Credit Limit vs Total Online Visits

- Customers with higher online visits generally tend to have higher average credit limits.
- Customers with low online activity mostly have lower to moderate credit limit.
- Several high credit limit customers show strong online banking engagement with frequent online visits.
- Customers with higher online interaction may prefer digital banking services over traditional banking methods.
- The relationship indicates that digitally active customers are likely to be financially active customers as well.



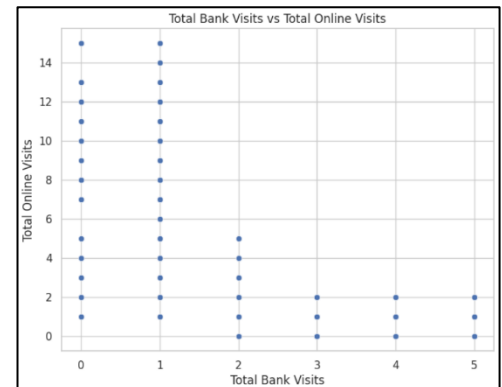
4.6.4 Average Credit Limit vs Total Calls Made

- Customers with fewer customer service calls show a wider range of average credit limits.
- Several high credit limit customers make very few customer support calls.
- Customers making a higher number of calls generally have lower credit limits.
- High call frequency may indicate greater dependency on customer support services.
- The relationship suggests that high-value customers rely less on customer service interactions.



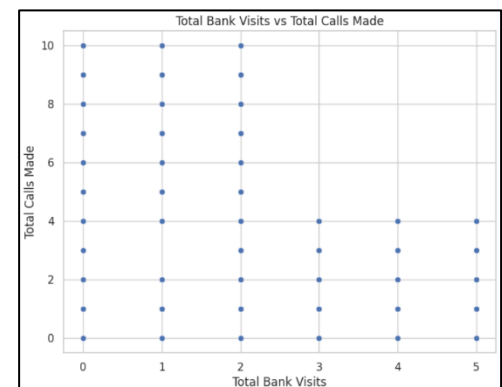
4.6.5 Total Bank Visits vs Total Online Visits

- Customers with lower bank visits generally show higher online visit activity.
- Customers who frequently visit the bank tend to have lower online banking visits.
- Several customers with 0 or 1 bank visits have very high online interaction.
- The relationship indicates different customer preferences between physical banking and digital banking channels.
- Customers appear to prefer either online banking services or physical branch visits rather than actively using both equally.



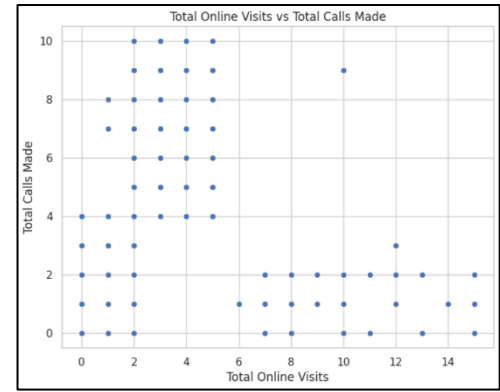
4.6.6 Total Bank Visits vs Total Calls Made

- Customers with lower bank visits show a wider range in the number of customer service calls.
- Customers making frequent bank visits generally make fewer customer support calls.
- Higher customer support interactions are mostly observed among customers with 0 to 2 bank visits.
- The relationship suggests that some customers prefer resolving issues through customer support rather than visiting bank branches.
- Different customers show different preferences for banking interaction channels such as branch visits and support calls.

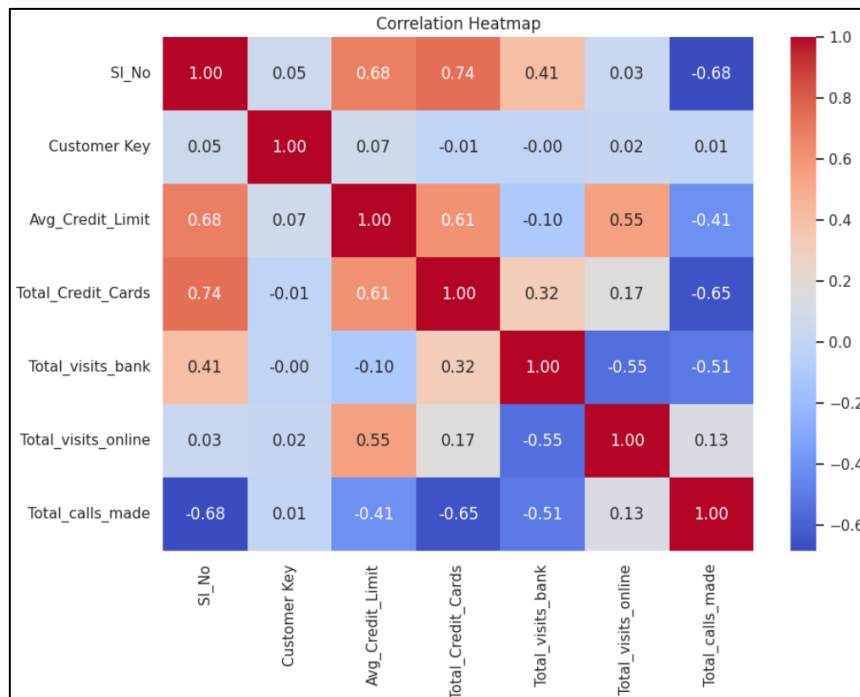


4.6.7 Total Visits Online vs Total Calls Made

- Customers with lower online visits tend to make a higher number of customer service calls.
- Customers with higher online banking activity generally make fewer customer support calls.
- Most highly digital customers have low call frequency, indicating smoother banking experience or self-service usage.
- Higher customer support dependency is mainly observed among customers with low online engagement.
- The relationship suggests that increased digital banking usage may reduce dependency on customer support services.



4.7 Correlation Analysis



- *Avg_Credit_Limit* shows a positive correlation with *Total_Credit_Cards* and *Total_visits_online*, indicating that customers with higher credit limits tend to own more credit cards and use online banking more frequently.
- *Total_visits_bank* has a negative correlation with *Total_visits_online*, suggesting that customers who frequently use online banking tend to visit bank branches less often.
- *Total_calls_made* shows a negative correlation with *Avg_Credit_Limit* and *Total_Credit_Cards*, indicating that customers with higher financial engagement make fewer customer support calls.
- *Customer Key* has almost no correlation with other variables since it acts only as an identifier.
- The correlation patterns indicate clear differences in customer banking behavior, making the dataset suitable for customer segmentation analysis.

5. Data Preprocessing

5.1 Checking Missing Values

```
Sl_No      0
Customer Key 0
Avg_Credit_Limit 0
Total_Credit_Cards 0
Total_visits_bank 0
Total_visits_online 0
Total_calls_made 0
dtype: int64
```

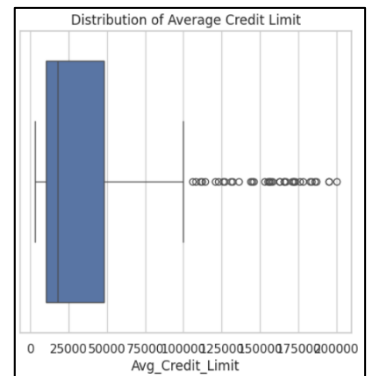
- All variables contain 0 missing values, indicating that the dataset is complete.
- No null values are present in any column of the dataset.
- The absence of missing values reduces the need for additional data cleaning or imputation techniques.
- The dataset is clean and suitable for further analysis and clustering techniques.

5.2 Outlier Detection

5.2.1 Visual Inspection using Boxplots

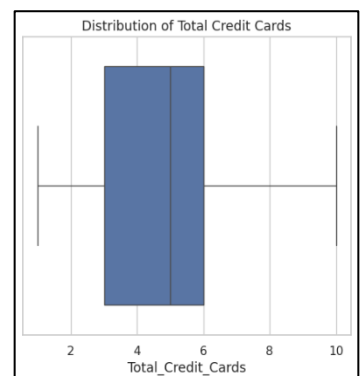
1. Average Credit Limit

- The boxplot shows the presence of multiple outliers in the *Avg_Credit_Limit* variable.
- Most customers have lower to moderate credit limits, while a smaller number of customers have very high credit limits.
- The skewness value of **2.20** indicates that the distribution is positively skewed.
- The high-value credit limit customers contribute to the long upper tail of the distribution.
- These outliers may represent premium or high-value customer segments and can provide important insights during customer segmentation analysis.



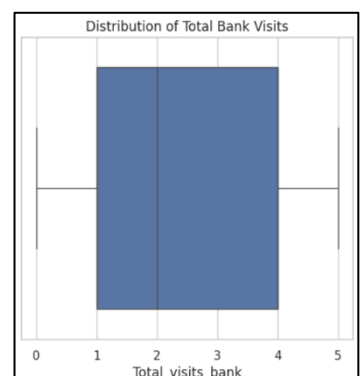
2. Total Credit Cards

- The distribution of *Total_Credit_Cards* is fairly balanced across customers.
- Most customers own between 3 to 6 credit cards.
- The median number of credit cards is around 5.
- No significant outliers are observed in the variable.
- The skewness value of **0.14** indicates that the distribution is nearly symmetric.
- The variable shows moderate variation in customer credit card ownership patterns.



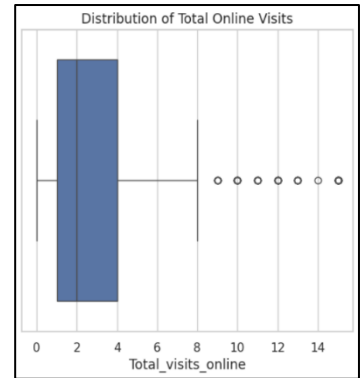
3. Total Bank Visits

- Most customers visit the bank branch between 1 to 4 times.
- The median number of bank visits is around 2.
- No significant outliers are observed in the variable.
- The skewness value of **0.14** indicates that the distribution is nearly symmetric.
- The variable shows moderate variation in customers' branch-visit behavior.



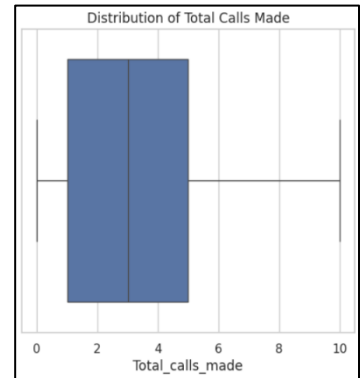
4. Total Online Visits

- Most customers have between 1 to 4 online visits.
- A smaller group of customers shows very high online banking activity with visits ranging from 9 to 15.
- Multiple outliers are present in the variable, representing highly active digital banking customers.
- The skewness value of **2.23** indicates that the distribution is positively skewed.
- The variation in online visits reflects differences in customer preference for digital banking services.



5. Total Calls Made

- Most customers make between 1 to 5 customer service calls.
- The median number of calls made is around 3.
- No significant outliers are observed in the variable.
- The skewness value of **0.66** indicates a slight positive skewness in the distribution.
- The variable shows moderate variation in customer support interaction behavior.



5.2.2 Detecting Outliers using IQR Method

```
Avg_Credit_Limit
-----
Lower Limit : -47000.0
Upper Limit : 105000.0
Number of Outliers : 39

Total_Credit_Cards
-----
Lower Limit : -1.5
Upper Limit : 10.5
Number of Outliers : 0

Total_visits_bank
-----
Lower Limit : -3.5
Upper Limit : 8.5
Number of Outliers : 0

Total_visits_online
-----
Lower Limit : -3.5
Upper Limit : 8.5
Number of Outliers : 37

Total_calls_made
-----
Lower Limit : -5.0
Upper Limit : 11.0
Number of Outliers : 0
```

- *Avg_Credit_Limit* contains **39 outliers**, indicating the presence of customers with exceptionally high credit limits.
- *Total_visits_online* contains **37 outliers**, representing customers with very high online banking activity.
- No outliers are observed in *Total_Credit_Cards*, *Total_visits_bank*, and *Total_calls_made*.
- The presence of outliers in credit limit and online visits suggests the existence of distinct customer groups with unique banking behavior.
- Since these outliers may represent important customer segments rather than data errors, they can provide valuable insights during clustering analysis.

5.3 Treating Outliers

- Outlier treatment was performed on *Avg_Credit_Limit* and *Total_visits_online* using the IQR method.
- Extreme values beyond the upper and lower limits were capped to reduce the impact of outliers.
- The treatment helps in minimizing the influence of unusually high values during clustering analysis.
- Only variables containing significant outliers were treated, while other variables remained unchanged.
- The overall structure and important customer behavior patterns in the dataset were preserved after outlier treatment.
- The dataset became more suitable for applying clustering algorithms and improving model performance.

5.4 Feature Engineering

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	\
0	100000.0	2	1	
1	50000.0	3	0	
2	50000.0	7	1	
3	30000.0	5	1	
4	100000.0	6	0	
	Total_visits_online	Total_calls_made		
0	1.0	0		
1	8.5	9		
2	3.0	4		
3	1.0	4		
4	8.5	3		

- The preview of the treated dataset shows that outlier handling was successfully applied to the selected variables.
- Extreme values in *Total_visits_online* have been capped within the acceptable range, with the maximum treated value observed as **8.5**.
- Other variables such as *Avg_Credit_Limit*, *Total_Credit_Cards*, *Total_visits_bank*, and *Total_calls_made* remain properly structured after treatment.
- The dataset retains important customer behavior information while reducing the impact of extreme values.
- The treated dataset is now more balanced and suitable for clustering and segmentation analysis.

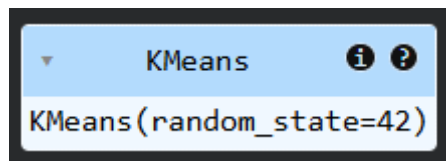
5.5 Data Scaling

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	\
0	2.398942	-1.249225	-0.860451	
1	0.643619	-0.787585	-1.473731	
2	0.643619	1.058973	-0.860451	
3	-0.058511	0.135694	-0.860451	
4	2.398942	0.597334	-1.473731	
	Total_visits_online	Total_calls_made		
0	-0.619767	-1.251537		
1	2.705813	1.891859		
2	0.267054	0.145528		
3	-0.619767	0.145528		
4	2.705813	-0.203739		

- The dataset has been successfully standardized using feature scaling techniques.
- The transformed values are centered around 0, with both positive and negative standardized values observed.
- Variables with different ranges have been converted to a common scale, ensuring fair contribution during clustering analysis.
- Standardization helps prevent variables with larger numerical values, such as *Avg_Credit_Limit*, from dominating the clustering process.
- The scaled dataset is now suitable for applying distance-based clustering algorithms such as K-Means and Hierarchical Clustering.

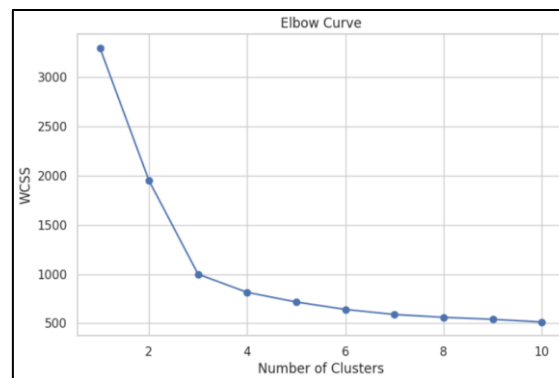
6. K-Means Clustering

6.1 Applying K-Means Clustering



- The K-Means clustering algorithm was initialized with a fixed random state value of **42**.
- Setting a random state ensures that the clustering results remain consistent and reproducible across multiple executions.
- K-Means is an unsupervised machine learning algorithm used to group customers with similar behavioral patterns.
- The algorithm will use the scaled customer data to identify meaningful customer segments based on similarities in banking behavior and financial activity.

6.2 Plotting the Elbow Curve



- The Elbow Curve shows a sharp decrease in WCSS values from 1 to 3 clusters.
- After **3 clusters**, the reduction in WCSS becomes more gradual.
- The elbow point appears at **K = 3**, indicating that 3 clusters may be the optimal number of customer segments.
- Choosing 3 clusters helps balance cluster compactness and model simplicity.
- The result suggests that the customer data can be effectively grouped into three meaningful customer segments using K-Means clustering.

6.3 Checking Silhouette Scores

```
Silhouette Score for 2 clusters : 0.4201583924929177
Silhouette Score for 3 clusters : 0.5169867470092899
Silhouette Score for 4 clusters : 0.38147048130671246
Silhouette Score for 5 clusters : 0.3535307587712862
Silhouette Score for 6 clusters : 0.2715116941900658
Silhouette Score for 7 clusters : 0.26718402363353444
Silhouette Score for 8 clusters : 0.2583338689382579
Silhouette Score for 9 clusters : 0.2520790833584104
Silhouette Score for 10 clusters : 0.2430988931035112
```

- The highest Silhouette Score is obtained for 3 clusters with a score of approximately **0.517**.
- A higher Silhouette Score indicates better cluster separation and cohesion.
- The score decreases after 3 clusters, suggesting reduced clustering quality for higher cluster values.
- The result supports the Elbow Method observation that 3 clusters is the optimal number of customer segments.
- Therefore, K-Means clustering with 3 clusters provides the most appropriate segmentation for the customer dataset.

6.4 Appropriate Number of Clusters

- The Elbow Curve showed a clear elbow point at 3 clusters, where the reduction in the WCSS started becoming gradual.
- The Silhouette Score for 3 clusters was the highest among all tested cluster values with a score of approximately **0.517**.
- A higher Silhouette Score indicates better cluster separation and more compact customer groups.
- Both evaluation techniques consistently suggest that 3 clusters provide the most meaningful and optimal segmentation of customers.

Therefore, the final K-Means clustering model was built using 3 clusters.

6.5 Cluster Profiling

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	\
Cluster				
0	33782.383420	5.515544	3.489637	
1	102660.000000	8.740000	0.600000	
2	12174.107143	2.410714	0.933036	
	Total_visits_online	Total_calls_made		
Cluster				
0	0.981865	2.000000		
1	8.180000	1.080000		
2	3.546875	6.870536		

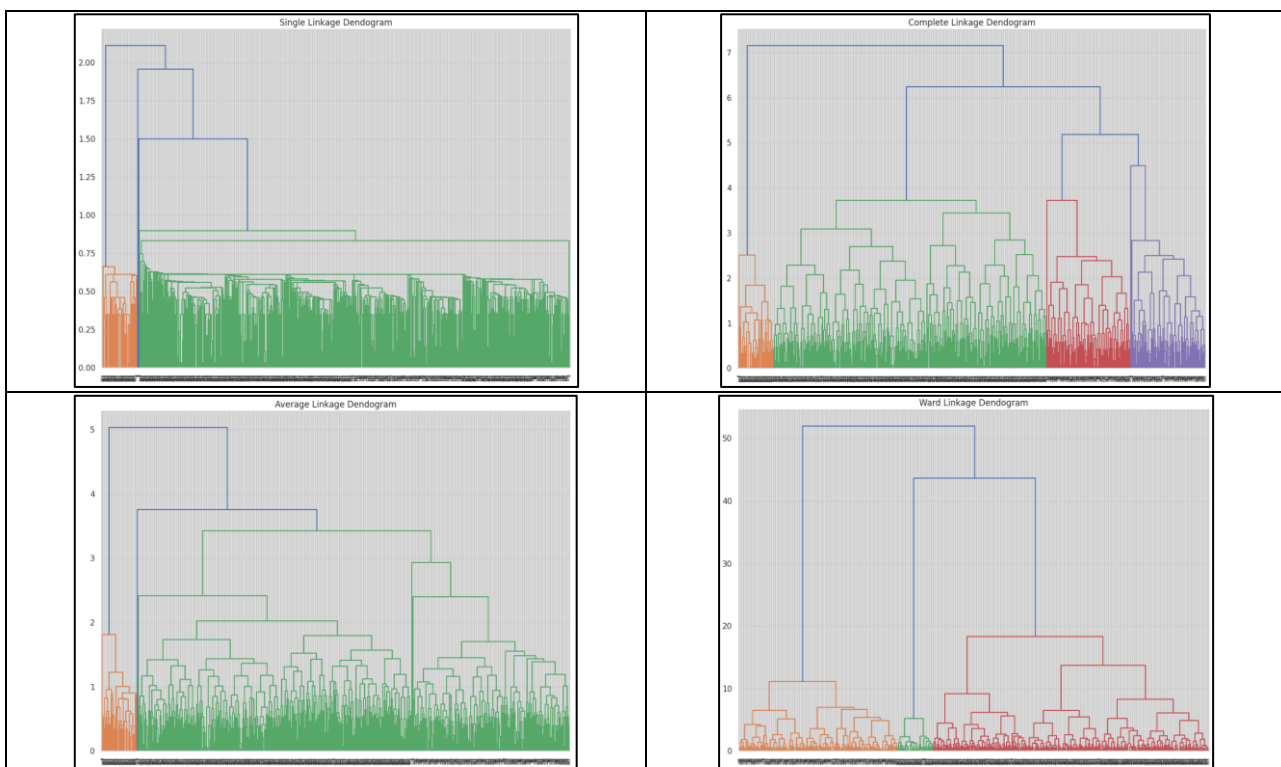
- **Cluster 0** consists of customers with moderate credit limits and a higher number of credit cards. These customers make moderate bank visits, low online visits, and relatively fewer customer service calls.
- **Cluster 1** represents high-value customers with the highest average credit limits and the most credit cards. These customers show very high online banking activity, minimal bank visits, and the lowest number of customer support calls.
- **Cluster 2** includes customers with the lowest average credit limits and fewer credit cards. These customers make fewer bank visits but comparatively higher customer service calls and moderate online visits.
- The clustering results clearly indicate distinct customer behavior patterns across the three customer groups.
- The identified clusters can help the bank design targeted marketing strategies and personalized banking services for different customer segments.

7. Hierarchical Clustering

7.1 Applying Hierarchical Clustering with Different Linkage Methods

- Hierarchical Clustering was successfully applied using four different linkage methods: Single, Complete, Average, and Ward linkage.
- Different linkage methods help analyze customer similarity using different cluster formation approaches.
- Single linkage forms clusters based on the minimum distance between observations.
- Complete linkage considers the maximum distance between observations while forming clusters.
- Average linkage uses the average distance between clusters for grouping customers.
- Ward linkage minimizes the variance within clusters and generally produces more compact and balanced clusters.
- Applying multiple linkage methods helps identify the most suitable hierarchical clustering approach for customer segmentation analysis.

7.2 Plotting Dendrograms for Different Linkage Methods



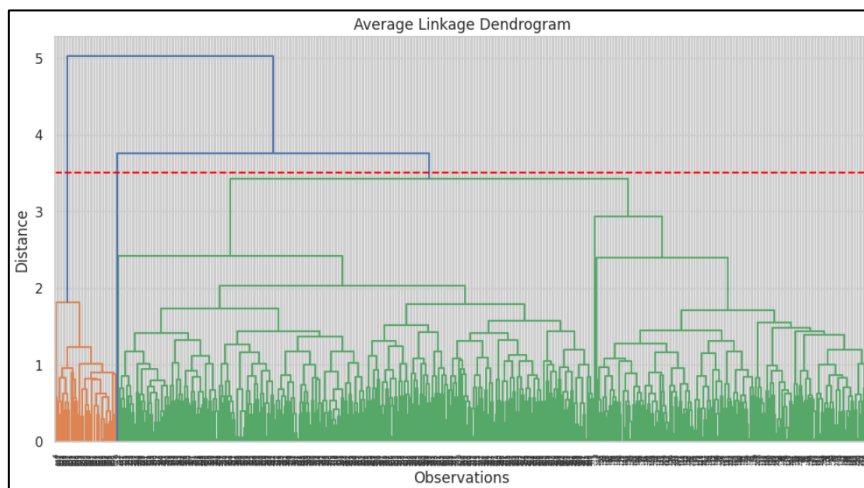
- The dendrograms generated using different linkage methods show different cluster formation patterns.
- The **Single Linkage** dendrogram shows a chaining effect, where clusters are merged gradually, making cluster separation less distinct.
- The **Complete Linkage** dendrogram forms more compact and clearly separated clusters compared to single linkage.
- The **Average Linkage** dendrogram provides balanced cluster grouping by considering the average distance between observations.
- The **Ward Linkage** dendrogram shows the clearest and most well-defined cluster separation among all methods.
- Ward linkage creates compact and homogeneous clusters by minimizing within-cluster variance.
- Based on the dendrogram analysis, **Ward Linkage** appears to be the most suitable hierarchical clustering method for customer segmentation.

7.3 Checking Cophenetic Correlation for Different Linkage Methods

```
Cophenetic Correlation for Single Linkage : 0.6666982080229052
Cophenetic Correlation for Complete Linkage : 0.8207884321476555
Cophenetic Correlation for Average Linkage : 0.8926672966587861
Cophenetic Correlation for Ward Linkage : 0.7702265240776373
```

- The Cophenetic Correlation Coefficient was calculated to evaluate the clustering quality of different linkage methods.
- **Complete Linkage** produced the highest cophenetic correlation value of approximately **0.821**, indicating the best preservation of pairwise distances among the methods tested.
- **Average Linkage** also showed a strong cophenetic correlation value of approximately **0.893**, indicating good clustering structure representation.
- **Ward Linkage** produced a reasonably high correlation value of approximately **0.770**, showing effective cluster formation.
- **Single Linkage** showed the lowest cophenetic correlation value of approximately **0.667**, indicating weaker clustering performance due to the chaining effect.
- Based on the cophenetic correlation results, **Average Linkage** and **Complete Linkage** provide better hierarchical clustering performance compared to the other methods.

7.4 Determining Appropriate Number of Clusters



- The Average Linkage dendrogram shows clear hierarchical grouping of customers based on similarity in banking behavior.
- A horizontal cut at the selected distance level forms approximately **3 major clusters**.
- The clusters appear reasonably balanced and well separated compared to single linkage clustering.
- The dendrogram indicates that customers within the same cluster share similar financial and banking activity patterns.
- Average Linkage provides better cluster compactness and separation by considering the average distance between observations.
- The clustering structure supports the formation of meaningful customer segments for further business analysis and targeting strategies.

7.5 Cluster Profiling

	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank	\
Cluster				
0	25725.779967	4.37931	2.553366	
1	102660.000000	8.74000	0.600000	
2	100000.000000	2.00000	1.000000	
	Total_visits_online	Total_calls_made		
Cluster				
0	1.925287	3.794745		
1	8.180000	1.080000		
2	1.000000	0.000000		

- **Cluster 0** consists of customers with moderate credit limits and moderate usage of banking services. These customers own around 4 to 5 credit cards and show balanced interaction through bank visits, online visits, and customer support calls.
- **Cluster 1** represents premium customers with the highest average credit limits and the highest number of credit cards. These customers are highly active in online banking, make very few bank visits, and require minimal customer support.
- **Cluster 2** includes customers with high credit limits but a very low number of credit cards. These customers show minimal online activity and almost no customer support interactions.
- The clusters clearly differentiate customers based on financial capacity, digital banking engagement, and service interaction behavior.
- The segmentation can help banks identify high-value digital customers, moderate banking users, and specialized customer groups for targeted business strategies and personalized services.

8. K-Means vs Hierarchical Clustering

K-Means Cluster Profile			
Cluster	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank \
0	33782.383420	5.515544	3.489637
1	102660.000000	8.740000	0.600000
2	12174.107143	2.410714	0.933036
Cluster	Total_visits_online	Total_calls_made	
0	0.981865	2.000000	
1	8.180000	1.080000	
2	3.546875	6.870536	

Hierarchical Clustering Profile			
Cluster	Avg_Credit_Limit	Total_Credit_Cards	Total_visits_bank \
0	25725.779967	4.37931	2.553366
1	102660.000000	8.74000	0.600000
2	100000.000000	2.00000	1.000000
Cluster	Total_visits_online	Total_calls_made	
0	1.925287	3.794745	
1	8.180000	1.080000	
2	1.000000	0.000000	

Both K-Means Clustering and Hierarchical Clustering successfully segmented customers into meaningful groups based on their banking behavior, credit card usage, online activity, and customer service interactions. In both techniques, a clear premium customer segment was identified, consisting of customers with high average credit limits, a greater number of credit cards, strong online banking engagement, fewer branch visits, and minimal dependency on customer support services. This consistency across both models indicates the presence of a highly valuable and digitally active customer group within the dataset.

K-Means Clustering created more distinct and business-oriented customer segments. It clearly separated customers with low credit limits and high customer service dependency from customers with moderate banking behavior and premium digital users. The clusters generated through K-Means were easier to interpret and provided clearer differentiation in customer behavior patterns, making the model highly suitable for practical business decision-making and targeted marketing strategies.

Hierarchical Clustering, particularly using Average Linkage, also produced meaningful customer groups with balanced cluster structures. The technique was effective in identifying similarities among customers and forming compact clusters based on overall behavioral patterns. However, compared to K-Means, the customer groups formed through Hierarchical Clustering appeared slightly broader and less behavior-specific in certain cases.

From the overall comparison, K-Means Clustering provided better segmentation clarity, stronger interpretability, and more actionable customer groups for business applications. The model effectively distinguished premium customers, moderate banking users, and service-dependent customers, making it more suitable for customer targeting, personalized banking services, and strategic decision-making.

8.1 Comparing Clusters obtained from K-Means and Hierarchical Clustering

1. Cluster 0 Comparison

In K-Means Clustering, Cluster 0 represents customers with moderate credit limits, a relatively higher number of credit cards, and frequent bank visits. These customers show low online banking activity and make fewer customer support calls, indicating a preference for traditional banking channels and stable banking behavior.

In Hierarchical Clustering, Cluster 0 also represents moderate customers with balanced banking activity. However, compared to K-Means, these customers show slightly lower credit limits and fewer credit cards. Their online visits and customer support interactions are comparatively moderate, indicating a more evenly distributed banking behavior across different service channels.

Overall, K-Means Cluster 0 provides a more clearly defined segment of traditional banking customers, while Hierarchical Clustering forms a broader moderate-customer group with balanced service usage.

2. Cluster 1 Comparison

Both K-Means and Hierarchical Clustering identified Cluster 1 as the premium customer segment. Customers in this cluster possess the highest average credit limits and the highest number of credit cards among all groups.

These customers are highly active in online banking, make very few branch visits, and show minimal dependency on customer support services. Their behavior reflects strong digital engagement, financial strength, and independent banking habits.

The consistency of this cluster across both techniques indicates a highly reliable and clearly distinguishable high-value customer segment. This group is highly important for banks as it represents profitable and digitally advanced customers who can be targeted for premium financial products and personalized services.

3. Cluster 2 Comparison

In K-Means Clustering, Cluster 2 represents low-value customers with the lowest average credit limits and fewer credit cards. These customers make the highest number of customer support calls and show moderate online activity, indicating greater dependency on banking assistance and support services.

In Hierarchical Clustering, Cluster 2 differs significantly from the K-Means result. Instead of low-value customers, this cluster contains customers with high credit limits but very low credit card ownership and minimal customer support interaction. These customers also show very low online banking activity, suggesting a unique customer group with selective banking usage behavior.

This difference highlights how K-Means focuses more on separating customers based on behavioral intensity, while Hierarchical Clustering identifies unique relationship patterns among customers. As a result, K-Means provides a more business-focused low-value customer segment, whereas Hierarchical Clustering captures a niche customer category with distinct banking characteristics.

9. Actionable Insights & Recommendations

9.1 Actionable Insights

- The analysis identified a premium customer segment with high credit limits, multiple credit cards, and strong online banking engagement. These customers are financially valuable and digitally active, making them ideal targets for premium banking products, exclusive rewards programs, wealth management services, and personalized financial offerings.
- Customers with moderate credit limits and frequent branch visits indicate a preference for traditional banking channels. Banks can improve engagement with this segment by promoting digital banking awareness programs, assisted digital onboarding, and hybrid banking services that combine branch support with online convenience.
- A separate customer segment showed high dependency on customer support services along with lower credit limits and fewer credit cards. This indicates potential dissatisfaction, lack of product understanding, or operational difficulties. Banks should focus on improving customer support efficiency, simplifying service processes, and providing proactive assistance to improve customer experience and retention.
- Customers with high online activity generally showed lower dependency on customer support and branch visits. This suggests that digital banking platforms are effectively improving customer convenience and self-service capabilities. Therefore, continued investment in digital banking infrastructure, mobile applications, and online service quality can enhance operational efficiency and customer satisfaction.
- The clustering analysis highlights the importance of personalized marketing strategies instead of applying a uniform approach to all customers. Different customer groups exhibit distinct financial behaviors, banking preferences, and service expectations. Segment-specific marketing campaigns, product recommendations, and communication strategies can significantly improve customer engagement and business performance.
- The presence of high-value customers with low service dependency indicates opportunities for customer loyalty programs and long-term relationship management. Retaining these profitable customers should be a strategic priority for the bank through exclusive benefits, faster service support, and customized financial solutions.
- The analysis also demonstrates that customer segmentation can help banks optimize resource allocation across physical branches, customer support services, and digital platforms. Understanding customer behavior patterns enables better decision-making regarding operational planning, service improvement, and future business growth strategies.

9.2 Business Recommendations

- The bank should implement a customer segmentation-based marketing strategy instead of using a uniform approach for all customers. Since customers exhibit different financial behaviors and service preferences, personalized banking solutions can improve customer engagement, satisfaction, and profitability.
- Premium customers with high credit limits, multiple credit cards, and strong digital banking engagement should be targeted with exclusive financial products such as premium credit cards, investment opportunities, wealth management services, higher reward programs, and personalized offers. Retaining these high-value customers should be a major business priority.
- Customers who frequently rely on customer support services should be provided with improved assistance mechanisms, faster issue resolution systems, and proactive customer relationship management. Banks can introduce dedicated support channels, simplified service processes, and educational guidance to reduce customer frustration and improve overall service experience.

- Since many customers prefer online banking over physical branch visits, the bank should continue investing in digital banking infrastructure, mobile banking applications, cybersecurity, and self-service platforms. Enhancing digital banking convenience can reduce operational costs while improving customer satisfaction and efficiency.
- Customers with moderate banking behavior and frequent branch visits can be encouraged to adopt digital banking services through awareness campaigns, digital literacy programs, assisted onboarding, and incentives for online transactions. This can help improve digital adoption while reducing dependency on branch operations.
- The bank should use clustering insights to improve cross-selling and upselling opportunities. Customers with moderate credit limits and fewer financial products can be targeted with suitable credit card upgrades, loan offers, insurance products, and customized financial plans based on their behavioral patterns.
- Resource allocation across customer support services, branch operations, and digital platforms should be optimized according to customer segment behavior. For example, high digital users may require fewer branch resources, while service-dependent customers may require stronger customer relationship management support.
- Regular customer segmentation analysis should be conducted to monitor changing customer behavior and market trends. Updating customer segments periodically can help the bank maintain effective marketing strategies, improve service quality, and strengthen long-term customer relationships.
- The insights generated from clustering analysis can also support strategic business decisions related to customer retention, operational efficiency, product development, and targeted promotional campaigns. By leveraging data-driven customer segmentation, the bank can enhance profitability, improve customer experience, and achieve sustainable business growth.

10. Conclusion

The objective of this project was to perform customer segmentation analysis using clustering techniques in order to identify distinct customer groups based on their banking behavior and financial activity. Through detailed exploratory data analysis, significant variations were observed in customer credit limits, credit card ownership, online banking usage, branch visits, and customer support interactions. These behavioral differences highlighted the need for customer segmentation to support better business decision-making and personalized banking strategies.

Both K-Means Clustering and Hierarchical Clustering were applied to the dataset to identify meaningful customer groups. The analysis revealed the presence of three major customer segments, including premium digitally active customers, moderate banking users, and customers with higher dependency on support services. Among the two techniques, K-Means Clustering provided more interpretable and business-oriented segmentation results, making it more suitable for practical business applications.

The findings of the analysis demonstrate that customer behavior varies significantly across different customer groups. Premium customers were found to have higher credit limits, multiple credit cards, strong online banking engagement, and minimal dependency on customer support services. In contrast, some customer groups showed greater reliance on branch visits and customer assistance, indicating different banking preferences and service requirements.

The project highlights the importance of customer segmentation in the banking industry. By identifying distinct customer groups, banks can design targeted marketing strategies, improve customer relationship management, optimize operational resources, and provide personalized financial services. The insights generated through clustering analysis can support data-driven business decisions that enhance customer satisfaction, improve retention, and increase overall profitability.

Overall, the project successfully demonstrated the effectiveness of clustering techniques in understanding customer behavior patterns and generating valuable business insights. The analysis provides a strong foundation for implementing customer-centric banking strategies and improving long-term business performance.